

Research Notes on Cross entropy loss

Cross entropy loss

$$H(p, q) = -\sum_x p(x) \log q(x)$$

>> Section 1

where $E p[\cdot]$ is the expected value operator with respect to the distribution p . The definition may be formulated using the Kullback–Leibler divergence $D_{KL}(p \parallel q)$, divergence of p from q (also known as the relative entropy of p with respect to q).

>> Section 2

$$y_i^{\wedge} = g(w \cdot x_i) = \frac{1}{1 + \exp(-(w_0 + w_1 x_{i1} + \dots + w_p x_{ip}))}$$

>> Section 3

$$\frac{\partial w_j}{\partial y_i^{\wedge}} = \frac{\partial w_j}{\partial \ln(1 + e^{-w_0 - w_1 x_{i1} - \dots - w_p x_{ip}})} = \frac{\partial w_j}{\partial (-w_0 - w_1 x_{i1} - \dots - w_p x_{ip})} = -x_{ij}$$

>> Section 4

Motivation In information theory, the Kraft–McMillan theorem establishes that any directly decodable coding scheme for coding a message to identify one value x_i out of a set of possibilities $\{x_1, \dots, x_n\}$ can be seen as representing an implicit probability distribution $q(x_i) = \frac{1}{2^{\ell_i}}$ over $\{x_1, \dots, x_n\}$, where ℓ_i is the length of the code for x_i in bits. Therefore, cross-entropy can be interpreted as the expected message-length per datum when a wrong distribution q is assumed while the data actually follows a distribution p . That is why the expectation is taken over the true probability distribution p and not q . Indeed the expected message-length under the true distribution p is

>> Section 5

The situation for continuous distributions is analogous. We have to assume that p and q

are absolutely continuous with respect to some reference measure r (usually r is a Lebesgue measure on a Borel σ -algebra). Let P and Q be probability density functions of p and q with respect to r . Then

>> Section 6

Put together, we obtain the desired result. Note that here we used the natural logarithm $\ln$ instead of $\log$ (canonically $\log_2$) in the loss function L . This merely changes the result by a factor of $\log(e)$ though. Moreover, for all i we reused $k_0 := -\sum_{l=1}^p w_l x_{il}$ and $k_j := -w_0 - \sum_{l \neq j} w_l x_{il}$ as constants w.r.t. each w_j .

>> Section 7

where we have used the calculation rules for the logarithm in the final line. Notice how the exponent contains a $-H(p, q_\theta)$ term. Taking the logarithm of both sides gives:

>> Section 8

$$-\int X P(x) \log Q(x) dx = E p[-\log Q],$$

>> Section 9

$$q_{y=1} = y^{\wedge} \equiv g(w \cdot x) = \frac{1}{1 + e^{-w \cdot x}},$$

>> Section 10

Cross-entropy minimization is frequently used in optimization and rare-event probability estimation. When comparing a distribution q against a fixed reference distribution p , cross-entropy and KL divergence are identical up to an additive constant (since p is fixed): According to the Gibbs' inequality, both take on their minimal values when $p = q$, which is 0 for KL divergence, and $H(p)$ for cross-entropy. In the engineering literature, the principle of minimizing KL divergence (Kullback's "Principle of Minimum Discrimination Information") is often called the Principle of Minimum Cross-Entropy (MCE), or Minxent. However, as discussed in the article Kullback–Leibler divergence, sometimes the distribution q is the fixed prior reference distribution, and the distribution p

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p is optimized to be as close to q as possible, subject to some constraint. In this case the two minimizations are not equivalent. This has led to some ambiguity in the literature, with some authors attempting to resolve the inconsistency by restating cross-entropy to be $D_{KL}(p \parallel q)$, rather than $H(p, q)$. In fact, cross-entropy is another name for relative entropy; see Cover and Thomas and Good. On the other hand, $H(p, q)$ does not agree with the literature and can be misleading.

>> Section 11

where $H(p)$ is the entropy of p . For discrete probability distributions p and q with the same support X , this means

>> Section 12

Since the logarithm is a monotonically increasing function, the maximizing value of θ is unaffected by this final step. Similarly, the maximizing value of θ is unaffected by the factor of N . So we observe that the likelihood maximization amounts to minimization of the cross-entropy.